

Subject Description Form

Subject Code	EIE4128
Subject Title	Integrated Capstone Project
Credit Value	6
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The objectives of this subject are to: • Provide students with the opportunity to apply and integrate the knowledge of artificial intelligence and data analytics (AIDA) to their discipline. • Develop the capabilities of students in analyzing and solving complex and possibly real-life problems using AIDA. • Train students with skills on systematic development and documentation of a significant piece of work.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Conduct literature surveys to locate materials and sources relevant to the selected problem area in a specific discipline. 2. Understand the materials obtained and connect the materials with the problem to be solved using AIDA knowledge and skills 3. Define and specify the problem precisely. 4. Assimilate and apply the learnt knowledge to generate good solutions to the problem. 5. Think critically the formulation of alternative models and solutions to the problem, in the analysis of approaches to the solution and their implementation. 6. Evaluate the final outcome in an objective manner. <u>Category B: Attributes for all-roundedness</u> 7. Improve presentation and communication skills via oral presentation. 8. Enhance technical report writing skills with proper organization of materials. 9. Develop the ability to learn independently and to find/integrate information from different sources required in solving real-life problems 10. Manage the project efficiently and effectively through the supervision of supervisor(s) 11. Work collaboratively with related parties (e.g. vendors, sponsor companies, technical support staff, team-partners, research students, etc.)

Subject Synopsis/ Indicative Syllabus	Students are expected to identify a project topic with a supervisor in their chosen discipline and a co-supervisor with artificial intelligence and data analytics expertise. Students need to demonstrate their knowledge in both their chosen discipline and AIDA in the project, receiving advice from both supervisors. The project should represent requisite effort in analyzing and interpreting the data/information obtained, using the principles and techniques learnt from various related subjects. Students are also expected to demonstrate significant analytical and, preferably, research ability in the chosen application domain. The progression of the project will consist of the following stages. <u>Project Specification</u> In this stage, the student will work in conjunction with the project supervisor to draw up a concrete project plan specifying at least the following: 1. Background of the project; 2. Aims and objectives; 3. Deliverables; 4. Methodology to be adopted; 5. Schedule. <u>Project Execution</u> After the specification is done, the project will be pursued so that the objectives are to be met; the deliverables are to be produced in accordance with the schedule. The student and the project supervisor will meet constantly to discuss the progress. In particular the following should be demonstrated: 1. Adherence to the schedule; 2. Achievement of objectives by the student's work; 3. Initiatives of the students to work, design, and to solve problems; 4. Inquisitiveness of the student (e.g. to probe into different phenomena or to try different approaches); 5. Diligence of the students to spend sufficient effort on the project; 6. Systematic documentation of data, design, results, etc. during the process of working out the project. <u>Project Report</u> After the project is completed, it is important that the student is competent in disseminating the results for others to review. Through this dissemination process, project achievements can be communicated, experience can be shared, knowledge and skills learnt can be retained and transferred. The following elements will be important as evidence of students' achievement: 1. Project logbook (documenting the work done over the year); 2. Project report (hardcopy and softcopy) 3. Presentation 4. Performance in a Question-and-Answer session; 5. Demonstration
Teaching/ Learning Methodology	The Integrated Capstone Project spans across the academic year for two consecutive semesters. The teaching/learning activities include regular project meetings with the supervisor and the co-supervisor, guided study of project materials, independent project development work and other project management tasks.

Assessment Methods in Alignment with Intended Subject Learning Outcomes	Specific Assessment Methods/ Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)											
			1	2	3	4	5	6	7	8	9	10	11	
	Continuous Assessment	100%	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
	Total	100 %												
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The assessment of the project work is conducted continuously throughout the whole project period. The evidence of students’ achievement will be documented in logbook and the reports submitted in various stages. The student will be required to give a presentation and demonstration so that he/she can communicate the project design, methodology, and achievement to other parties. The Integrated Capstone Project will be assessed by the supervisor/co-supervisor and other assessors. Attributes to be assessed include, but not limited to, Problem Identification, Problem Solving, Communication and Presentation, Project Management, and Self-Discipline. Integrated Capstone Projects should be problem-oriented and there are no restrictions on the nature of the problem except that it should be relevant to the student’s chosen discipline and AIDA. The project could be practical, academic or a hybrid in which the student is encouraged but not constrained to have some original contributions. Each student has to submit a proposal, a mid-term checkpoint progress report and a final report. The proposal must be approved by the supervisor/co-supervisor before the student can proceed with the Integrated Capstone Project. An oral presentation and demonstration are essential at the end of the project. A mid-term presentation and demonstration may also be required for proper continuous assessment.													
Student Study Effort Expected	Class contact (time-tabled):													
	• Structured Study (regular meetings with supervisor)											78 Hours		
	Other student study effort:													
	• Guided Study/Reading/Experiment • Reports • Presentation and Demonstration											132 Hours		
	Total student study effort:											210 Hours		
Reading List and References	Reference Books: To be specified by the project supervisor for each project.													